IND3001 Engineering Economics

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The role of Engineering in Capitalism

- Agricultural societies ---- Industrial Societies
- (Modernism-adaptation of societies moral issues)
- Subsistance production ------- Production for profit
- Use Value -------- Exchange value

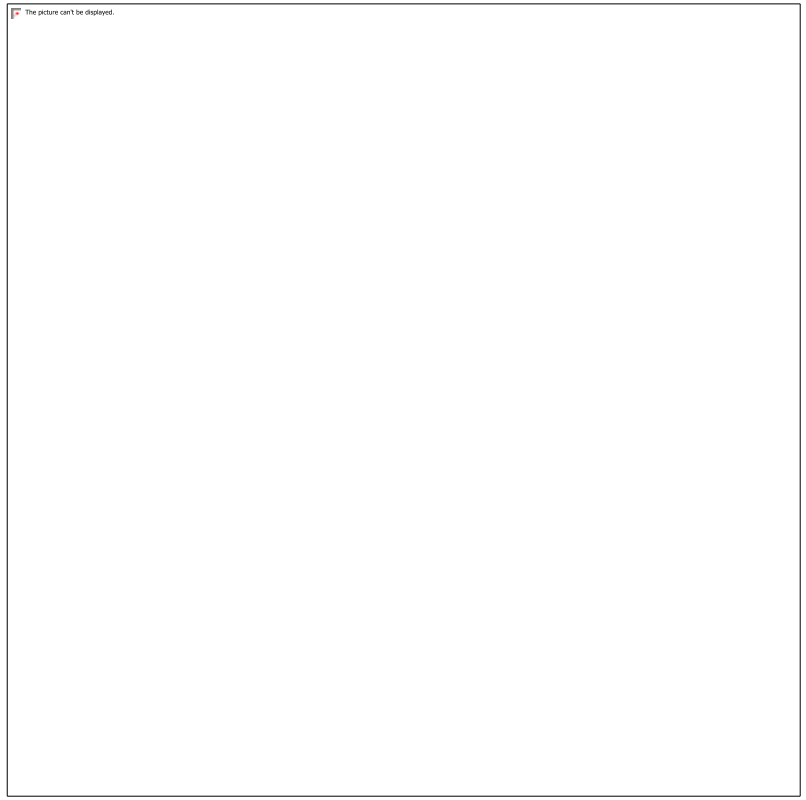
- Linear way to economic development in capitalism = Welfare

The Sources of Long-Run Growth

The Crucial Importance of Productivity

Rising Labour Productivity / Productivity /output per worker /shorten production time
Relative surplus value

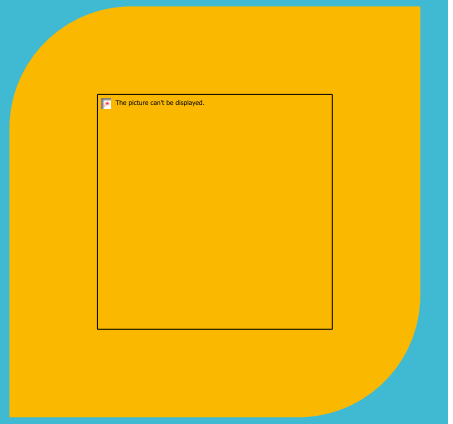
** For the economy as a whole, productivity is simply real GDP divided by the number of people working.



So increased productivity is the key to long-run economic growth.

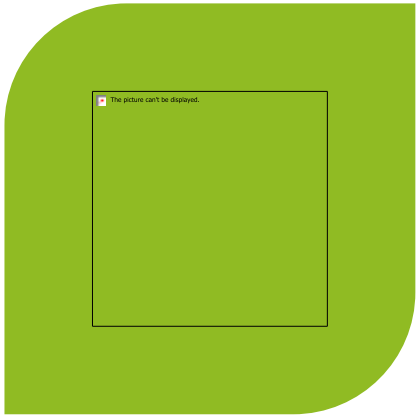
But what leads to higher productivity?





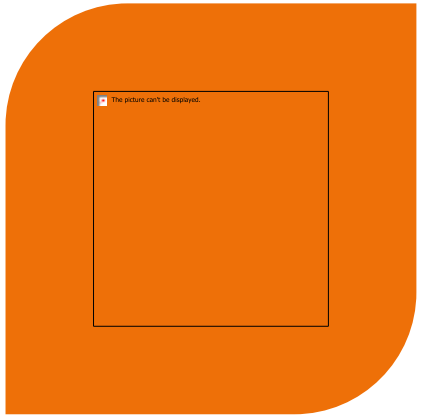
PHYSİCAL CAPİTAL

Physical capital consists of human-made resources such as buildings and machines.



HUMAN CAPİTAL

Human capital is the improvement in labor created by the education and knowledge embodied in the workforce.



TECHNOLOGY

Technology is the technical means for the production of goods and services.

Accounting for Growth: The Aggregate Production Function

- Productivity is higher, other things equal, when workers are equipped with more physical capital, more human capital, better technology, or any combination of the three.

  But can we put numbers to these effects? To do this, economists make use of estimates of the aggregate production function, which shows how productivity depends on the quantities of physical capital per worker and human capital per worker as well as the state of technology.

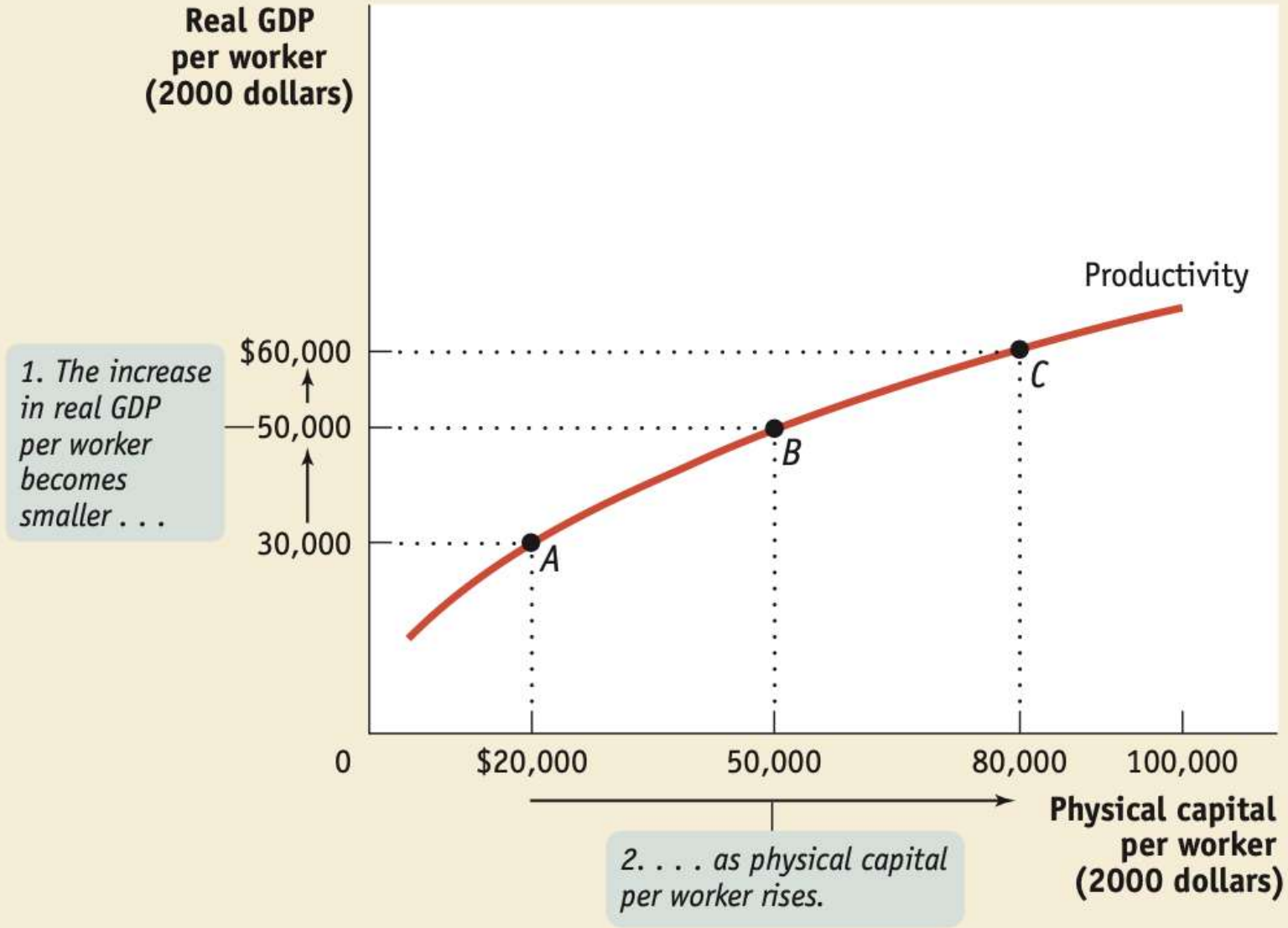


Aggregate production function: Y/L = f(K/L, H/L, T)

- f(•) means a function of the variables within the parentheses.
- Y is real GDP produced, and L is the number of workers employed, so that Y/L is real GDP produced per worker;
- K/L is the quantity of physical capital per worker;
- H/L is the quantity of human capital per worker;
- and T is a measure of the state of the technology used in production

Physical Capital and Productivity

Other things equal, a greater quantity of physical capital per worker leads to higher real GDP per worker, but is subject to diminishing returns: each successive addition to physical capital per worker produces a smaller increase in productivity. Starting at point *A*, with $20,000 in physical capital per worker, a $30,000 increase in physical capital per worker leads to an increase of $20,000 in real GDP per worker. At point *B*, with $50,000 in physical capital per worker, a $30,000 increase in physical capital per worker leads to an increase of only $10,000 in real GDP per worker. >web...



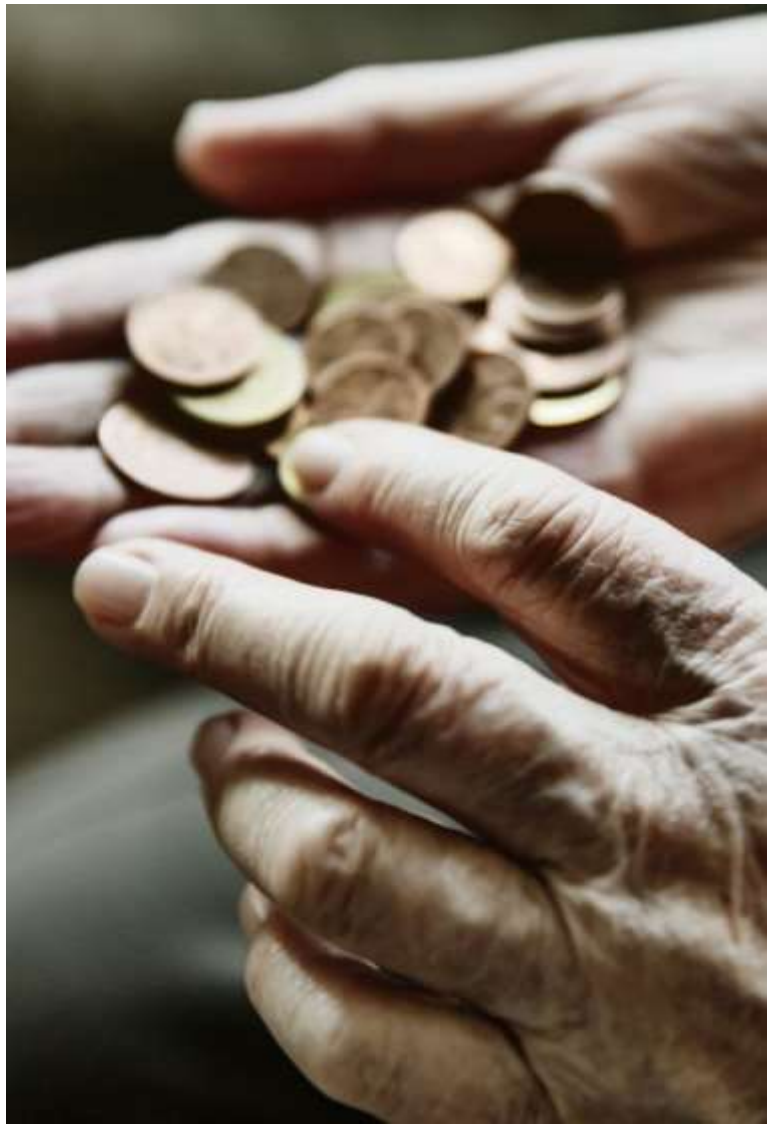
Why Growth Rates Differ?

- Even small differences in growth rates have large consequences over the long run. But why do growth rates differ across countries and across periods of time?
- The simplest answer is that economies with rapid growth tend to be economies that add physical capital, increase their human capital, experience rapid technological progress, or all three, on a sustained basis.
- The deeper answer lies in policies and institutions that promote economic growth—policies and institutions that ensure that those who generate additions to physical or human capital, or to technological progress, are rewarded for their efforts.



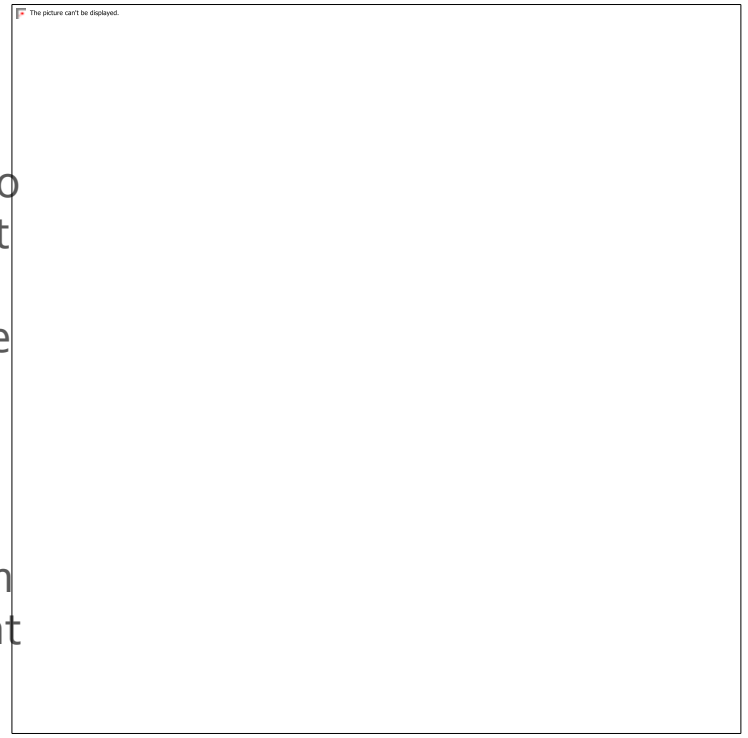
1)Savings and Investment Spending

- To increase the physical capital available to workers, an economy must engage in investment spending. There are two ways to do this:
- One way is for its residents to engage in saving.
  Such domestic savings can be generated by private households saving some of their disposable income, by the government spending less than its tax revenues, or both.
- The other source of funds for investment spending is foreign savings from residents of other countries.



2) Well-functioning financial system

- The degree of trust on financial and banking systems.
- A well-functioning financial system requires appropriate government regulation that assures depositors that their funds are protected.
- Government policy has an effect on savings and investment spending in two other important ways. The government can engage in saving by taking in more in taxes than it spends. By doing so, the government effectively makes more funds available for investment spending.
- Another way in which governments can adversely affect savings and investment spending is through irresponsible conduct of monetary policy—specifically, by generating excessive inflation



3) Foreign Investment

- A country's investment spending can be either more or less than its domestic savings because of foreign investment: residents of some countries invest part of their savings in other countries.
- Investors expect and normally receive a return on their investment. If a country's residents have borrowed from foreigners, they have to pay interest on those loans. And if foreign companies build new factories or purchase existing factories in a country, the foreign owners are entitled to receive any profits those factories earn.
- Still, economists argue that the increase in real GDP generated by foreign investment is usually greater than the interest and profits paid to foreigners. So, the country benefits overall from foreign investment.

4) Education

- An economy's physical capital is created mainly through investment spending by individuals and private companies. Much of an economy's human capital, in contrast, is the result of government spending on education.



5) Infrastructure

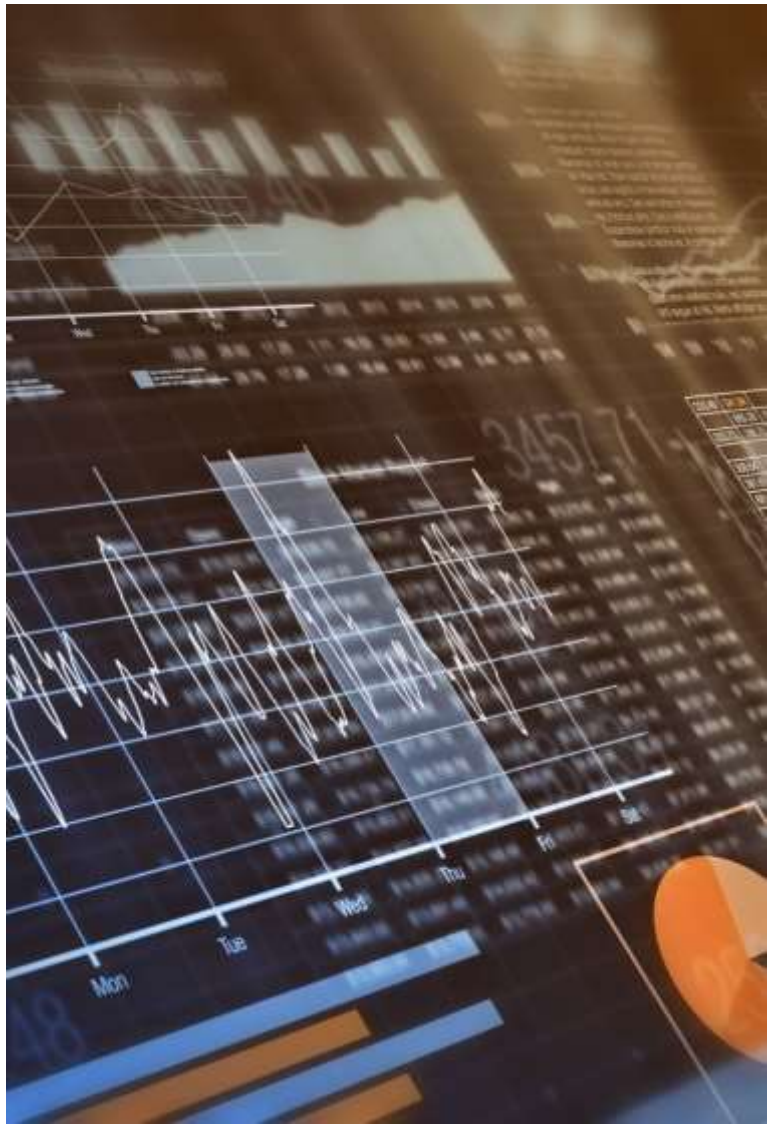
The term infrastructure refers to roads, power lines, ports, information networks, and other parts of an economy's physical capital that provide an underpinning, or foundation, for economic activity.

Although some infrastructure is provided by private companies, much of it is either provided by the government or requires a great deal of government regulation and support.



6) Research and Development

- The advance of technology is a key force behind economic growth. What drives technology?
- Scientific advances make new technologies possible.
- But science alone is not enough: scientific knowledge must be translated into useful products and processes.



- **Engineering economics** is the application of economic principles to engineering decisions.
- Engineers constantly face choices between technically feasible alternatives: which machine to buy, whether to repair or replace equipment, which design to build.
- Engineering economics provides the tools to choose the alternative that is **economically best**, not just technically possible.
- *Engineering asks 'Can we build it?'*
  Engineering economics asks 'Should we build it, and which way is best?'

- The founding text of the discipline is usually taken to be the 1887 book by the American railway engineer **Arthur Mellen Wellington**, *The Economic Theory of the Location of Railways*. Wellington's definition in the preface is still quoted
- engineering, he says, is **"the art of doing that well with one dollar which any bungler can do with two."**
- It contains the engineer's entire self-image: the engineer is *the one who saves*. Saves what? Material, time, energy and labour. When you say "one dollar instead of two," part of that dollar is almost always **somebody's wage**.

Time of engineering economics

- The time of engineering economics generally concerns **the future**. Consequently, the figures used in the discipline are the best estimates of what is expected to occur.
- Estimates and decisions relate to:

-Cash flows

-The timing of cash flows

-Interest rates associated with the time value of money

-Measuring the economic value associated with selecting an alternative

Core Concepts /Alternatives

- In engineering applications, it is often possible to perform a task in various ways. By comparing these options rationally, the most economical alternative can be selected.
- These alternatives generally involve factors such as purchase cost, annual operating and maintenance costs, useful life, interest rate, and salvage/scrap value.
- We evaluate these factors to determine which alternative is the most economical.

Core Concepts /Cash Flows

- Estimated cash inflows and outflows are referred to as cash flows.
- This constitutes the heart of the analysis and is often its most important component.
- However, it is, of course, impossible to know exactly what future oil and natural gas prices will be.

Types of Efficiency

- Efficiency of a system is generally defined as the ratio of its output to input. The efficiency can be classified into *technical efficiency* and *economic efficiency*.

- **Technical Efficiency**
- It is the ratio of the output to input of a physical system. The physical system may be a diesel engine, a machine working in a shop floor, a furnace, etc

$$\text{Technical efficiency } (\%) = \frac{\text{Output}}{\text{Input}} \times 100$$

The technical efficiency of a diesel engine is as follows:

$$\text{Technical efficiency } (\%) = \frac{\text{Heat equivalent of mechanical energy produced}}{\text{Heat equivalent of fuel used}} \times 100$$

- **Economic Efficiency**
- Economic efficiency is the ratio of output to input of a business system.
- Economic efficiency is also called 'productivity'. There are several ways of improving productivity.
- 1) Increased output for the same input
- 2) Decreased input for the same output
- 3) By a proportionate increase in the output which is more than the proportionate increase in the input
- 4) By a proportionate decrease in the input which is more than theproportionate decrease in the output
- 5) Through simultaneous increase in the output with decrease in the input.

$$\text{Economic efficiency } (\%) = \frac{\text{Output}}{\text{Input}} \times 100 = \frac{\text{Worth}}{\text{Cost}} \times 100$$

Taylorism/ Fordism

- Henry Ford (1863–1947) is the founder of Ford Motor Company. Introduced in 1908, the Model T was a simple, durable and standard car. Ford's aim was to turn the automobile from a luxury good for the rich into a mass product.

- In 1913, at the Highland Park plant, Ford introduced the moving assembly line. The worker no longer went near the piece, the piece passed in front of the worker. The results were striking:

- Assembly time for a Model T chassis dropped from approximately 12.5 hours to approximately 1.5 hours.
- The price of the Model T fell from about $850 in 1908 to about $260–$300 in the 1920s.
- More than 15 million Model T had been produced when production ceased in 1927.

Thanks!
Q&A

